

Aayush Shrestha

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EDUCATION

Morgan State University

B.S. in Computer Science

Baltimore, Maryland

Expected Graduation, May 2026

- **GPA:** 3.96/4.00, Dean's List
- **ExpandQISE Fund Scholar**, full-year NSF-funded tuition scholarship for academic excellence and AI research contributions

EXPERIENCE

Gaudon Silicone Sealants

Co-Founder & CTO

[GitHub]

Remote

Nov 2025 - Present

- Launched a B2B e-commerce platform using **React**, **Vite**, **Tailwind CSS**, and **GSAP**, serving **35+ wholesale clients** and **5 distributors** across the U.S. with a smart product finder, comparison tool, and automated quote system [gaudon.store]
- Automated lead generation with an agentic **skill** scraping **200+ wholesale leads**, sending personalized cold emails via **Gmail API** with **OAuth 2.0**, and syncing follow-ups to **Google Calendar**

Morgan State University

Research Assistant & Lead Developer - AI Agents

[GitHub]

Baltimore, Maryland

Jun 2025 - Present

- Architected a **9-agent RAG system** for the **NAO6** robot using **OpenAI**, **Pinecone**, and a custom **Python** DSP pipeline with voice activity detection for real-time speech transcription and response generation [GitHub]
- Building multi-agent advising systems with **Google ADK**, **Vertex AI**, and **LangChain** RAG pipelines on **Cloud Functions**, automating course planning and student support for **800+ CS students**

Ace Data Cloud

Software Engineering Intern

Remote

May 2024 - Aug 2024

- Built **Python/FastAPI** API wrappers for external AI model integrations with **Redis** caching, reducing average API response latency by **20%**; containerized services with **Docker** Compose and deployed to production on **AWS** with automated CI/CD pipelines

PROJECTS

Morgan State University AI Chatbot (CS Navigator)

Lead Developer

[GitHub]

Baltimore, Maryland

2026

- Led a team of 4 building a full-stack **Retrieval-Augmented Generation (RAG)** web app with **React 19**, **FastAPI**, **LangChain**, **Pinecone**, and **OpenAI GPT-3.5** for semantic search over university data
- Implemented **JSON Web Token (JWT)** authentication with role-based session management, multi-turn chat history with context persistence, voice input/output, and a **DegreeWorks PDF parser** for automated academic transcript imports
- Deployed on **AWS EC2** with **RDS MySQL**, **Docker**, and **Nginx** reverse proxy; serving **800+ CS students** at Morgan State [inavigator.ai]

Agentic Skills Suite (Open Source)

Creator & Developer

[GitHub]

Remote

2025 - 2026

- Created **8 autonomous AI agent skills** (Gmail, Calendar, Lead Gen, Twitter Brief, NotebookLM) powered by **MCP protocol**, shared **Google OAuth 2.0** auth, **Gemini API**, and **Playwright** browser automation
- Shipped an **apply-job** skill with **120+ installs** on skills.sh automating end-to-end job applications via **Playwright MCP**, handling login flows, form filling, resume uploads, and recruiter outreach

Selected Projects

Developer

Remote

2025 - 2026

- Engineered an autonomous AI agent using the **ReAct reasoning pattern** with **Google Gemini** to monitor and respond to **Twitter** mentions with context-aware replies in real time; deployed as a serverless application on **Vercel** with **Twitter API v2** [GitHub]
- Fine-tuned **TrOCR** and **BERT** models to automate metadata extraction from historical manuscripts, extending the model context window to **3,000 tokens** using **PyTorch** and the **Hugging Face Transformers** library for long-document processing [Colab]

ACTIVITIES & LEADERSHIP

- **Presenter**, AAAS Annual Meeting 2026 (Phoenix, AZ). Presented research on integrating **OpenAI GPT** models with **NAO6** robots for real-time conversational AI using **Whisper**, **Pinecone RAG**, and gesture synthesis; evaluated by a panel of three judges
- **Speaker**, EdgeCon Winter 2026 (Princeton, NJ). Showcased CS Navigator alongside **Dr. Paul Wang**, Chair of Computer Science at Morgan State University, to industry professionals and higher education technology leaders
- **Selected Presenter**, Amazon AI/ML HBCU Symposium (Arlington, VA, Sept 2025). Delivered Generative AI robotics research to **AWS** engineers and ML scientists at Amazon HQ2; featured in an AWS [technical podcast interview] on generative AI and robotics
- **Teaching Assistant**, COSC 220, Data Structures & Algorithms under **Dr. Jin Guo** @MSU. Mentoring 30+ students through weekly office hours and code reviews; developing supplementary problem sets on linked lists, stacks, queues, trees, and graphs

SKILLS

Programming: Python, JavaScript, TypeScript, Java, Kotlin, SQL, R, HTML/CSS

Frameworks & Tools: React, Next.js, Vite, Tailwind CSS, FastAPI, Flask, Node.js, LangChain, PyTorch, Docker, Nginx, Git

Cloud & Databases: AWS (EC2, S3, Lambda, RDS), Google Cloud (Vertex AI, ADK), Pinecone, PostgreSQL, MySQL, MongoDB, Redis

AI/ML & Other: LLMs, RAG Systems, AI Agents, MCP, Claude Code, Vector Databases, NLP, REST APIs, CI/CD, Microservices